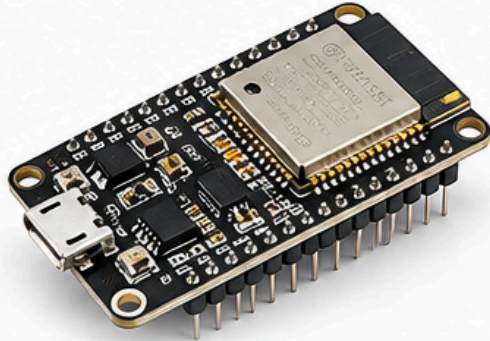


AGROTRACE COOL[🌱] — COMPONENT LIST

YUVA YODHA ENERGY TECH HACKATHON • CHALLENGE 01 • SUSTAINABLE AGRICULTURE

1 ESP32 Microcontroller



Reads sensors, runs decision logic, controls fan & water pump

2 DHT22 Sensor



Measures ambient temperature and relative humidity

3 DS18B20 Probe



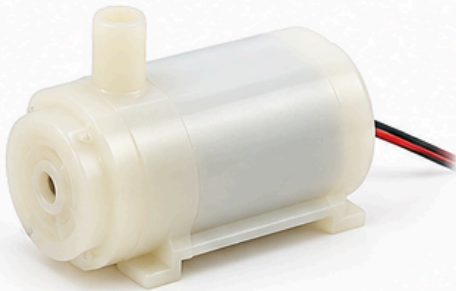
Waterproof probe — measures produce/core temperature

4 Small DC Fan



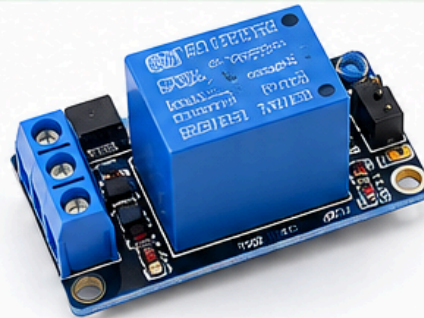
Draws air through the wet cooling pad

5 Small DC Water Pump



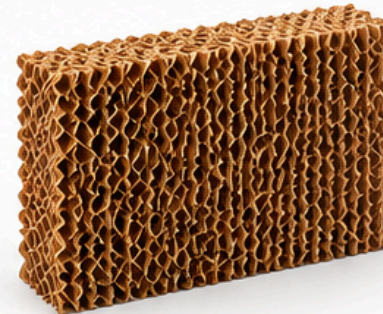
Supplies water to the evaporative cooling pad

6 Relay / MOSFET Driver



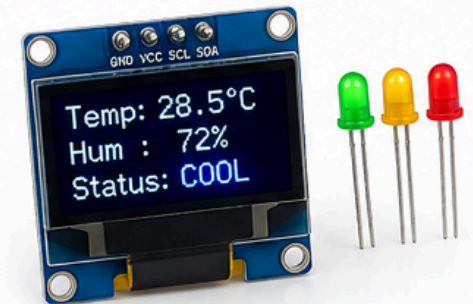
Safely switches fan & pump — ESP32 never drives them directly

7 Wet Cooling Medium



Jute / coir pad — where evaporative cooling takes place

8 OLED Display & Status LED



Shows live status on-device — COOL / REDUCE / STOP

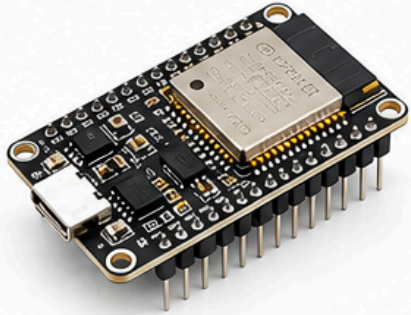


SENSE • ANALYZE • COOL • ADVISE • RECORD

AGROTRACE COOL[🌱] — SOFTWARE STACK

YUVA YODHA ENERGY TECH HACKATHON • CHALLENGE 01 • SUSTAINABLE AGRICULTURE

1 ESP32 Firmware



Reads sensors, runs decision logic, logs data and communicates with the Android app

2 Android Application



Operator dashboard for monitoring, batch management and real-time status

3 Data Communication



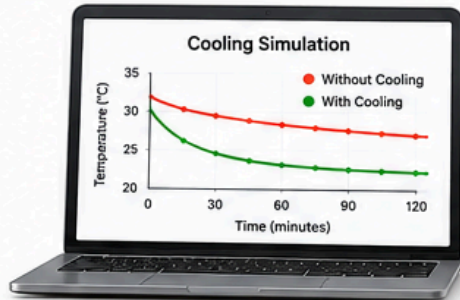
ESP32 and Android app communicate via Bluetooth (ESP32 BLE) for real-time data sync

4 Local Data Storage



Sensor data and cooling logs are stored locally on a microSD card in the ESP32 module

5 Cooling Simulation



Simulation module to demonstrate cooling behavior under different environmental conditions

6 Data Processing & Logic



Python scripts for data processing, cooling decision logic and simulation analysis (used for development and testing)

7 Data Visualization



Graphs and batch-wise data to help operators understand cooling performance

8 App Data Storage



Batch information and sensor data saved in the Android app (SQLite) and can be exported as CSV for records



SENSE • ANALYZE • COOL • ADVISE • RECORD